1st File Continue…..

Mid Term Current File

ZOO103

Effort by CH Ismail

REMEMBER ME IN YOUR PRAYERS

Q. Define Pedipalps and Epigyne? 2

Pedipalps (commonly shortened to palps or palpi) are the second pair of appendages of chelicerates.

Epigyne or epigynum is the external genital structure of female spiders

.Q. Name of Ecological Variations? 5

1) Habitat Variation.
2) Variations induced by climatic conditions.
3) Host determined variations.
4) Density dependent variations.
5) Allometric variations.
6) Neurogenic color variations

Q. Explain Five Kingdom classifications? 5

Five kingdoms proposed by Robert Whittaker include:

• Monera

• Protista

• Fungi

• Plantae

• Animalia

Q. Courtship behavior in jumping spider?

When a male jumping spider encounters a female—literally any female—he launches into an elaborate courtship dance, including rhythmic flailing of limbs and complex vibrations. (Read "Bondage, Cannibalism, and Castration—Spiders' Wild Sex Lives.")

Q. Non-probability sampling method?

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Non-probability sampling is a method of selecting units from a population using a subjective (i.e. non-random) method. Since non-probability sampling does not require a complete survey frame, it is a fast, easy and inexpensive way of obtaining data.

Q. Ecological character's name?

Each species has specific niche

• Types of niches

• Habitats and hosts environment

• Food sources

• Seasonal variation

• Parasite and host interaction reaction

Q. Components of systematics?

Component fields of systematics• Component fields of systematics are: • Biodiversity• Taxonomy• Classification• Nomenclature• Biogeography• Evolutionary biology

Q. Random sampling?

If every individual of target population has an equal chance of selection then it will be called random sampling.

Q. What Is Stratified Random Sampling?

Stratified random sampling is a method of sampling that involves the division of a population into smaller sub-groups known as strata.

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Q. Achaea and eubacteria difference?

Characteristics	Eubacteria	Archaea
Cell type	Prokaryotic	Prokaryotic
Cell wall	Contain peptidoglycan	Carbohydrates, proteins or No cell wall
Interons	No interons	Interons present
Ribosomal proteins	No ribosomal proteins	Ribosomal proteins present
Diseases	Can be parasitic and cause diseases	None are parasitic
Gene resemblance	No resemblance	Eukaryotes

Q. Prosoma? 2

The anterior region of the body of an invertebrate (such as an arachnid) especially when the segmentation is suppressed or obscured especially: cephalothorax.

Q. Evolutionary species concept? 3

• An evolutionary species "is a single lineage of ancestor descendant populations of organisms which maintains its:

• Identity from other such lineages [in space and time]

Q. Probability sampling types? 5

➢ Simple Random Sampling.

➢ Stratified Random Sampling.

➢ Systematic Random Sampling.

➢ Cluster (Area) Random Sampling.

➢ Multi-Stage Sampling.

Q. Social variation in ants? 2

• Ants live in complex social colonies.

Queen: founder and leader of the colony.

• Primary goal is to populate colony. • Lay thousands of eggs

Queen Chambers are deep within the colony for protection. • Long life in some species up to 30 years.

Drones: Only function is to fertilize female • Die after fertilization • rarely encountered outside the colony

Workers: Foraging and protection • Sometime: queen lay eggs, winged male and females-Alates or reproductive.

Q What do you now name different eyes in spider species? 3

Most spiders have eight eyes. • Some species of spiders, such as those that live in caves or under the soil, have no eyes at all. • 99% of all spiders have 8 eyes • 1% nearly all have 6, but there are a few exceptions. • Varying number of eyes in the same spider family.

What are three domain systems? 3

- Archaea
- Eubacteria
- Eukaryota

Q. What is evolution?

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Evolution is a process by which plants and animals that exist today have been evolved from plants and animal of past.

Q. Methods of taxonomic key?

• Read introductory comments, format of the key, abbreviations etc. before using key

• Read both choices at each step of the key

• Use glossary to find out the definition of any term your do not understood

• Use a ruler if measurements are needed, does not guess.

Q. Feeding habits in jumping spiders? 5

Jumping spiders eat all kinds of insects, and have been known to kill grasshoppers that dwarf them in size. To aid in their hunting, jumping spiders have large, powerful front legs that they use to hold the prey in place. Unlike other kinds of spiders, jumping spiders do not use webs to catch their prey.

Q. Neurogenic variation in ecological variation? 5

• It is color change in response to environment.

• Chameleons have Chromatophores in three layers below their transparent skin.

• Xenthophors (Yellow) and erythrophores (red) in upper layer.

• Iridophores or guanophores (blue or white) 2nd layer.

• Melanophores (dark colour) in deep layer.

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Q. Why arthropods are preserved? 3

• Arthropods are easy to process as killed immediately and preserved and store in alcohol

• Crabs and prawn may be killed in formalin but renders their joints hard and brittle.

• Larger invertebrate need to be injected 10% formalin to prevent from rotting

• Industrial alcohol is used for most arthropods.

Q. Why members of the species resemble? 2

1. When organisms breed they shuffle their gene pool which gives the species its identity.

2. Genes are not shared with other species so they look different

Q. Problems of typological species concept? 3

• Polymorphism within populations.

• Sibling or cryptic species.

• Geographical variations among populations

Q. Courtship rituals in lynx spider? 5

In a typical mating sequence of the green lynx spider, courtship lasted 11 minutes. Mating took 10 minutes, consisting of several copulations measured from the first approach of a male palpus to the female's epigynum until the final separa- tion.

Q. Legs of arthropods? 3

Four pairs of legs/Chelicera/padipalp

Legs are covered with many hairs

Claws 2-3 at the end of legs

Parts of Legs

• Coxa • Traochanter • Femur • Patella • Tibia • Metatarsus • Tarsus • Claws

Q. Internal morphology?

• Internal anatomy provides Taxonomic Characters for all groups of higher animals.

• The extend to use vary from group to group.

• The skull of routinely used in classification of higher mammals.

• Teeth and bones of fossils and other hard parts.

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Q. Theoretical biology?

• Evolution

• Ecology

• Molecular Biology

• Evolutionary Biology

Q. Venom gland?

Parts of Venom gland and use

• Poison gland

• Poison Duct

• Fang

• Absent in the family Uloboridae. The glands are located either in the chelicerae or under the carapace.

Q. Post-Mortem Variations?

Chinese jay plumage changes from green to blue in collection • Wasps over-expressed to cyanide turn bright red • Insects • Spiders

Q. Reproduction in spider?

• Male finds a female

• A male spider deposits his sperm into a sperm web, holding it in his palps.

• Female spiders produce either one egg sac containing several to a thousand eggs or several egg sacs each with successively fewer eggs. Females of many species die after producing the last egg sac.

Q. Four lineage of Arthropod?

1) Trilobites – extinct
2) Chelicerates – horseshoe crabs, spiders
3) Uniramians – centipedes, millipedes, insects
4) Crustaceans – crabs, lobsters, barnacles

Q. Holotype?

If no specimen designated or indicated as the type by the original author but it is clear that only one specimen was used for description then it will be considered as Holotype.

Q. Sibling Species?

Two or more than two closely related species which are morphologically alike but behaviorally or reproductively isolated from each other.

Examples are Drosophila

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Q. Define systematics?

• Scientific study of kinds and diversity of organisms and of any or all relationship among them • Broad term which include Taxonomy, Nomenclature and Classification etc.

Q. Electrophoretic difference?

• This method determines the difference in the isozyme and allozyme.

• Can help to determine share character among taxa.

Q. Three basic characteristics of arthropods?

1) Some are aquatic, others are terrestrial, some are free living, and other is parasitic.
2) All arthropods have a body supported by hard exoskeleton.
3) Legs, antennae and mouth are jointed.
4) Tube like heart.
5) Overall body bilateral symmetrical

Q. Monotypic and polytypic species?

Monotypic Species

Species with a single subspecies, called monotypic species

Example: Bardala labarda

Polytypic Species:

A species contains two or more subspecies

Examples are tiger, Panthera tigris which has several subspecies

Q. Physiological characters?

Physiological characters are low weight characters and are variable:

• All structures are product of physiological process and are thus physiological characters.

• This character cannot be studied in preserved animals.

• Metabolic activities

• Body secretion

• Genic sterility factor

Q. Any five morphological characters?

Morphology describes the structural features. Morphological characters include:

1. External morphology.
2. Genetelic structure
3. Internal morphology
4. Embryology
5. Karyology and other cytological differences

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Q. Hunting spider?

Hunting spiders typically have only two claws.

Q. Define habitat variation? 3

Populations of single species in different habitat are visibly different.

Oyster in still water

Round or broader

Oyster in strong currents

Narrow, elongated and usually straight

Q. Sympatric speciation?

• A speciation in which a species is evolved from a single common ancestral species while inhabiting the same geographical range.

• Share the same habitat but become reproductively isolated from each other.

• More common in plants.

Q. Peripatric speciation?

• Peripatric speciation is a mode of speciation in which a new species is formed from an isolated peripheral population.

• Resemble alloprtraic speciation/barriers.

• Isolating population is small.

Example:

An example of this is the London Underground mosquito

Q. Allometric variation?

• Allometry results in misappropriate size of some structure in relation to rest of the body.

• This results in animals of different sizes----Allometric variability

Q. Genetic variation?

• Difference in the individual members of a population

• Genetic variations are important for survival and adaptation of species

• Evolution• Natural selection

Causes of evolution:

• Mutations

• Genetic recombination

• Genetic drift

• Migration

Q. Seasonal variation?

• Same animal may have different appearance in different parts of the year

• In arctic and sub-arctic birds there may be change from white winter dress to normal colour summer dress.

Q. Traumatic variation? 5

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• Teratology is the study of abnormalities of physical development.

• Taratogen is the toxic substances which cause abnormal development.

• Mercury, alcohol, lead, thalidomide • Teratological variations are common in animals.

Q. Biology and biochemistry difference?

Q. What are the aims of systematics? 5

• Invent world's kinds of organisms (flora and fauna)

• Provide method for identification and communication

• To produce coherent and universal system of classification and

• Demonstrate evolutionary implications of biodiversity

Q. Biological drawbacks?

• Asexual reproducing organisms.

• Fossils

• Sibling or cryptic species

• Distinguishing between species on the basis of reproductive separation is problematic because reproductive isolation is often incomplete.

Q. Applied biology?

• Agriculture and forestry

• Biological control

• Public health

• Quarantine

• Wild life management

• Mineral prospecting

• National defense

• Environmental problem

• Soil fertility

• In commerce

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Q. Wet method?

• Wet method of preservation involves the use of liquid preservatives to prevent deterioration.

• Chemicals like alcohols, formalin etc. are commonly used for specimen preservation. Animals belonging to phylum porifera, annelida, nematodaare mostly preserved by wet method.

Q. DNA hybridization:

• This method is used to determine the genetic distance between two organisms

• Mainly measure the genetic similarity.

Q. Types of Non-Genetic variations?

• Individual variations in Time

• Social Variations

• Ecological Variations

• Traumatic variation

• Post-morten variations

Q. Catalogue of life?

• The Catalogue of Life is an online database • It provides the most comprehensive and authoritative index of known species of organisms. • The Catalogue interface is available in twelve languages and is used by research scientists, citizen scientists, educators, and policy makers.

Q. Upward and downward taxonomy? 5

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Best of luck

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